

Kitchen Environment

breakdown of the **coordinates and dimensions** for all major items in the kitchen Unified Robot Description Format (URDF) model, organized by component. This includes positions (XYZ), orientations (RPY), and geometric properties (size/scale) for each item.

1. Table Area

- **Link:** table_area
 - **Position (XYZ):** (-2.7, -0.6, 0) (relative to room_link)
 - **Orientation (RPY):** (0, 0, π) (180° rotation around Z-axis)
 - **Mesh:** big_table_1.dae (visual), big_table_1.stl (collision)
 - **Dimensions** (approximate, inferred from mesh):
 - **Width:** ~1.5m (X-axis)
 - **Depth:** ~0.8m (Y-axis)
 - **Height:** ~0.75m (Z-axis)
 - **Center:** (0, 0, 0.375) (local coordinates)

2. Sink Area

Sink Structure

- **Link:** sink_area
 - **Position (XYZ):** (0.29, 1.03, 0.41) (relative to sink_area_footprint)
 - **Mesh:** SinkArea.dae
 - **Collision Box** (for surface):
 - **Size:** (0.575, 1.6, 0.02) (X, Y, Z)
 - **Center:** (0, -0.225, 0) (local coordinates)

Sink Bowl

- **Link:** sink_area_sink
 - **Position (XYZ):** (0.005, 0.47, 0.4414) (relative to sink_area)
 - **Mesh:** Sink.dae
 - **Dimensions:** Inferred from mesh (no explicit size in URDF).

Drawers (Example: Trash Drawer)

- **Link:** sink_area_trash_drawer_main
 - **Joint:** Prismatic (sliding along X-axis)
 - **Position (XYZ):** (0.01, 0.7, -0.015)
 - **Limits:** lower="0" upper="0.48" (48cm travel)
 - **Mesh:** Drawer_60_58.dae
 - **Handle Position:** (0.3325, 0, 0.2375) (relative to drawer).

3. Oven Area

Oven Structure

- **Link:** oven_area_oven_main
 - **Position (XYZ):** (0.0, 0.0, 0.415) (relative to oven_area_area)
 - **Mesh:** OvenMain.dae
 - **Door Joint:** Revolute (Z-axis)
 - **Position:** (0.2737, 0.0, -0.2614)
 - **Limits:** lower="0" upper="0" (initially locked).

Oven Knobs

- **Example Knob** (oven_area_oven_knob_stove_1):
 - **Position (XYZ):** (0, -0.21, 0) (relative to oven_area_oven_panel)
 - **Orientation (RPY):** (0, $-\pi/2$, 0) (90° around Y-axis)
 - **Mesh:** Knob.dae
 - **Rotation Limits:** 0 to π (180°).

4. Kitchen Island

Island Structure

- **Link:** kitchen_island
 - **Position (XYZ):** (0.2975, 1.1292, 0.421) (relative to kitchen_island_footprint)
 - **Mesh:** IslandArea.dae
 - **Surface Collision Box:**
 - **Size:** (0.8, 2.45, 0.02) (X, Y, Z)
 - **Center:** (0, 0, 0.42) (local coordinates).

Stove

- **Link:** kitchen_island_stove
 - **Position (XYZ):** (0.0, 0.7658, 0.4315)
 - **Mesh:** Stove.dae.

Drawers (Example: Left Upper Drawer)

- **Link:** kitchen_island_left_upper_drawer_main
 - **Joint Position (XYZ):** (0.0025, -0.8, 0.114)
 - **Mesh:** Drawer_60_29.dae
 - **Handle Position:** (0.3325, 0, 0.11).

5. Fridge Area

Fridge Structure

- **Link:** iai_fridge_main
 - **Position (XYZ):** (0, 0, 0) (relative to fridge_area)
 - **Mesh:** Fridge.dae
 - **Door Joint:** Revolute (Z-axis)
 - **Position:** (0.29, 0.3, 0.245)
 - **Limits:** 0 to $\pi/2$ (90°).

Fridge Door Handle

- **Link:** iai_fridge_door_handle
 - **Position (XYZ):** (0.06, -0.57, 0.0)
 - **Mesh:** VHandle90.dae.

6. Global Coordinates Summary

Component	Parent Link	Position (XYZ)	Orientation (RPY)	Dimensions (W×D×H)
Table	room_link	(-2.7, -0.6, 0)	(0, 0, π)	~1.5m × 0.8m × 0.75m
Sink Area	room_link	(1.825, 1.32, 0)	(0, 0, π)	Collision: 0.575×1.6×0.02
Oven Area	room_link	(1.805, 2.52, 0)	(0, 0, π)	Mesh-dependent

Kitchen Island	room_link	(-1.365, 0.59, 0)	(0, 0, 0)	Surface: 0.8×2.45×0.02
Fridge Area	room_link	(1.825, -0.74, 0)	(0, 0, π)	Mesh-dependent

Key Notes:

1. **Positions** are relative to the parent link (e.g., sink_area is child of sink_area_footprint).
2. **Dimensions** are either explicit (collision boxes) or inferred from meshes.
3. **Orientation**: RPY = Roll-Pitch-Yaw (in radians). $\pi = 180^\circ$.
4. **Handles/Drawers**: Prismatic joints define sliding ranges (e.g., 0.48m max for drawers).